Basic information theory

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Information theory, which is an important application area of probability theory, was originally invented to describe communication channels. It was brought to wide attention through an article by *Claude Shannon* in 1948. Information theory has been widely applied to brain theory. For example, some researchers have attempted to quantify the information in spike trains with formal methods of information theory, and the study of the 'neural code' was popular in the late 1990s. Concepts of information theory are used in this book at several places, including the discussion of coding in neural maps and the formulation of the anticipating brain. The following is a basic introduction to some of the constructs invented in this theory.

D.1 Communication channel and information gain

One can talk very casually about information and how information is transmitted by signals over telegraph wires or in neural systems. However, it is useful to define information more formally so that we can quantify the amount of information that is or can be transmitted in different systems. This was done in 1948 by *Claude Shannon* who studied the transmission of information over a general *communication channel* as shown in Fig. D.1. A message x_i, one of several possible messages of a set X from an information source, is converted into a signal $s_i = f(x_i)$ that can be transmitted over the communication channel to a receiver. The receiver converts the received signal r_i back to a message $y_i = g(r_i)$, one of a set Y of possible output messages, that can be interpreted by the destination receiver. Shannon included a noise input η in the channel that is intended to capture faulty transmissions, for example due to physical noise in the transmission channel or faulty message conversion. In the noiseless case the receiver receives the sent signal $r_i = s_i$, which can be converted into the original message when the receiver does the inverse function of the encoding done by the transmitter, that is, $g = f^{-1}$. Note that we can consider several noise models such as additive noise, for example, $r = s + \eta$, or multiplicative noise, for example, $r_i = s_i * \eta$, where η is a random variable.

What is the amount of information we gain when receiving a message y_i? This depends highly on our expectations. For example, if I tell you that the first name of 'Shannon' is 'Claude' then you gain no information if you have read carefully the text above where it was already mentioned. If you thought it was either 'Albert' or 'Claude' you gain some information. You gain more information from my message if you thought it was one of four possible choices

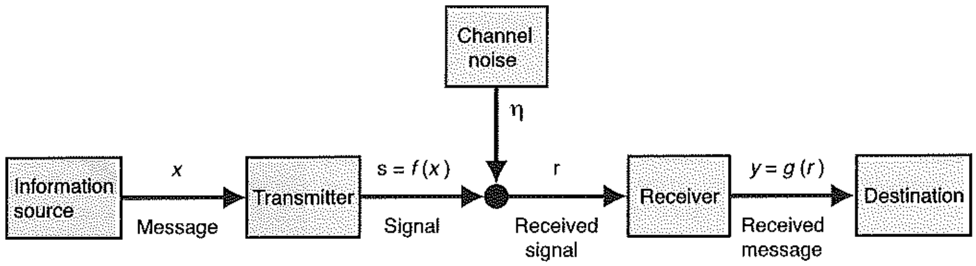

Fig. D.1 The communication channel as studied by Shannon. A message x is converted into a signal $s = f(x)$ by a transmitter that sends this signal subsequently to the receiver. The receiver generally receives a distorted signal r consisting of the sent signal s convoluted by noise η. The received signal is then converted into a message $y = g(r)$. [Adapted from Shannon, *The Bell System Technical Journal* 27: 379–423 (1948).]

with equal likelihood. Information depends on the set of possible messages and their frequencies, and a measure of information gain should therefore be a function of the probability $p_i = P(y_i)$ of a message. Furthermore, independent information should be additive, that is, if I tell you that the middle name of Claude Shannon is 'Elwood', then you gain a bit of information independent of and additional to the information I gave you previously. The probability of independent messages is the product of the probabilities of the individual messages. We thus need a function (of the probabilities) that has the property

$$f(xy) = f(x) + f(y). \tag{D.1}$$

The logarithm has these characteristics, and we therefore define the information gain when receiving a message y_i as

$$I(y_i) = -\log_2(p_i). \tag{D.2}$$

The minus sign makes the information positive as the probabilities are less than 1. Note that we use the logarithm with basis 2 and no additional proportionality constant, which is just a convention. This convention defines the units of one *bit*. If we have two possible states with equal likelihood, then the transmission of one state allows us to gain 1 bit of information.

Let us illustrate the concept further with information transmitted by hand signals as illustrated in Fig. D.2. In Fig. D.2A we illustrate a case where we use only the thumb in two possible locations, either horizontal or up, to transmit a message, say '0' and '1'. If both cases are equally likely, that is, $P(0) = P(1) = 1/2$, then we receive an information gain of

$$I = -\log_2(0.5) = 1 \tag{D.3}$$

bit of information. If we use two fingers with two possible locations and the code of the total number of fingers to transmit information, as illustrated in Fig. D.2B, then we can transmit three different messages. The gain of information from each signal is therefore $I = -\log_2(1/3) = 1.585$ bits with equally frequent signals. In the code illustrated in Fig. D.2B it does not matter whether the pointing finger is extended or if the thumb is up. In contrast, in Fig. D.2C

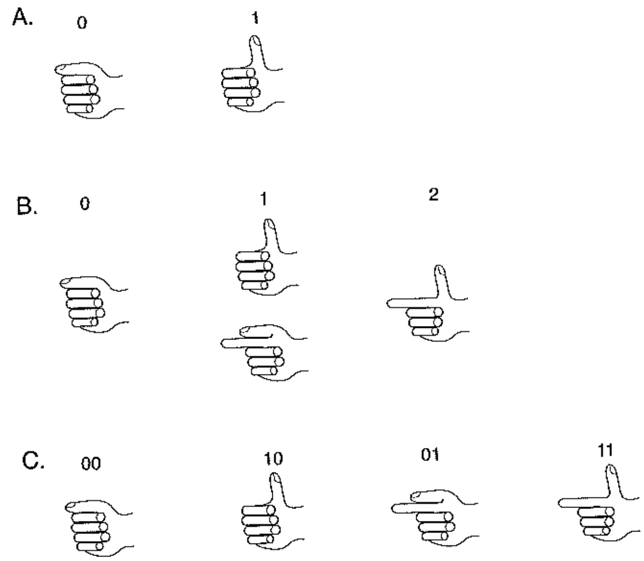

Fig. D.2 Example of hand signals used to transmit messages. (A) Only two positions of the thumb are used to encode two possible messages '0' or '1'. (B) Two fingers are used to encode three messages by the total number of extended fingers. (C) Two fingers are used with a different code, in which the position of each of the two fingers used for the signalling is essential.

we have changed the code to transmit a message with two fingers where now the position of the finger matters. We can then transmit four different messages, and each equally likely signal would convey 2 bits of information.

We demonstrated the information gain with discrete events that had a certain probability distribution. Discrete events are also most appropriate when thinking about message transmission with a finite alphabet. However, it is mathematically often more convenient to write the following formulations in continuous form. The information gain is thus often written in the form

$$I(x) = -\log_2 P(x). \tag{D.4}$$

Note that this formulation can still be used for discrete events. The probability function $P(x)$ then has non-vanishing values only for a set of discrete values x. If we consider a continuous set X of events x then we have to take the corresponding probability density function $p(x)$ into account (see also Appendix C). The probability of an event x is then an infinitesimally small quantity $P(x) = p(x)\mathrm{d}x$, and the information gain for one of those events goes to infinity. However, averages of continuous sets are still defined as we will see in the following, and it is easy to replace integrals with sums (and vice versa) in the formulas below.

We have so far only considered the cases where a message conveyed a definitive statement. In contrast, if I tell you that the first name of Shannon is 'Claude' with 90% probability out of four possibilities, then there is still some uncertainty left which we have to take into account. We have then to distinguish the probability of an event before the message was sent, called the *a priori* or *prior* probability $P^{\text{prior}}(x)$, and the probability of an event after taking the information into account, which is called the *posterior* probability

$P^{\text{posterior}}(x)$. The information gain is then defined by

$$I(x) = -\sum_x P^{\text{posterior}}(x) \log_2 \frac{P^{\text{prior}}(x)}{P^{\text{posterior}}(x)}. \tag{D.5}$$

In the previous examples we knew the precise answer after the message was received, so that $P^{\text{posterior}}(x) = 1$. We also wrote the prior probability simply as $P^{\text{prior}}(x) = P(x)$. Equation D.4 is therefore only a special case of the more general formula, eqn D.5.

D.2 Entropy, the average information gain

We have used equally likely signals in the previous numerical examples. It is, however, often the case that not all messages are equally likely, and the amount of information gain is therefore different for different messages. For example, let's go back to the example illustrated in Fig. D.2A and consider the case where the thumb is up in 3/4 of all messages sent out and only horizontal in 1/4 of the cases. When we receive a signal of thumb horizontal, then we gain $I(0) = -\log_2(1/4) = 2$ bits of information, whereas the signal of 'thumb up' conveys only $I(1) = -\log_2(3/4) = 0.415$ bits of information. The average amount of information in the message set of the information source is

$$S(X) = -\sum_i p_i \log_2(p_i), \tag{D.6}$$

which is called the *entropy* of the information source. The entropy is a quantity of the message set and is not defined for an individual message. The entropy of the set of possible received messages $S(Y)$ is defined in a similar way. The average amount of information gained by the transmission of a signal in the previous example is $S = 1/4*2+3/4*0.415 = 0.8113$. The entropy is a measure of the average information gain we can expect from a signal in a communication channel, though it is good to keep in mind that individual messages can have larger and smaller information gain values. Some rare events might indeed convey a very large amount of information.

The entropy of a message set with N possible messages that are all equally likely is

$$S(X) = -\sum_{i=1}^{N} \frac{1}{N} \log_2(\frac{1}{N}) = \log_2(N). \tag{D.7}$$

The entropy is hence equivalent to the logarithm of the number of possible states for equally likely states. This is a useful fact to keep in mind. More generally, the entropy measures the effective number of states by weighting the logarithm of the number of states with the frequency of their occurrences.

We can formulate the entropy for continuous distributions in an analogous way. We again write the set with capital letters, for example, X, and examples from this set with lower case letters, for example, x. The entropy of a set X with probability density function $p(x)$ is then defined as

$$S(X) = -\int_X p(x) \log_2 p(x) \mathrm{d}x. \tag{D.8}$$

For example, the entropy of a Gaussian-distributed set with mean μ and variance σ^2,

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{D.9}$$

is given by

$$S = \frac{1}{2}\log_2(2\pi e\sigma^2). \tag{D.10}$$

This entropy does depends on the variance, not the mean. This is intuitively clear as the average information gain is only a relative quantity and information is transmitted by variations of a signal around its mean value.

D.2.1 Difficulties in measuring entropy

Measuring entropies is a data hungry task. This comes partly from the fact that we have to estimate probability distributions. To do this we have to observe many instances, from which we can, for example, construct a histogram. The histogram gives us the numbers of the frequencies with which events occur within a certain resolution determined by the bin size of the histogram. We can often get some reasonable estimations of the probability function if we have at least some prior knowledge about the form of the probability function. We can then fit the histograms with the expected function to fix the unknown parameters to values consistent with the experimental observations. Often, however, we do not know *a priori* the form of the distribution functions, and suitable guesses from the data are common. We have then to check that the results do not depend critically on the choice of distribution function.

A further difficulty is that information depends on the logarithm of probabilities. This means that events with small probabilities have a large factor in the entropy, and reliable measurements of rare events do demand a large amount of representative data. To get probability distributions from frequency histograms we have to normalize the histograms so that the sum over all data becomes one (see Appendix C),

$$\int \mathrm{d}x P(x) = 1. \tag{D.11}$$

Due to this normalization procedure we tend to overestimate the information content of the events measured if we miss out some rare events with high information content. Estimations of entropy therefore tend to overestimate the true entropy. This is a potential danger for suitable interpretations of experimental results. For example, a large entropy per spike, much larger than, for example, 2 bits, would rule out rate codes and would therefore indicate that temporal structures in the spike trains are used to transmit information. An overestimation of entropies from experimental data can obscure such conclusions. Some methods have been developed to compensate for such systematic shifts of small data sets, but such methods cannot, of course, completely compensate for real data.

D.2.2 Entropy of a spike train with temporal coding

It is very instructive, and important for the comparisons with experimental data outlined below, to calculate the maximum entropy of a spike train with temporal coding up to a certain precision. To do this we introduce small time bins Δt, small enough that one time bin can contain at most one spike. We can then view the spike train as a binary string with 0 in a time bin corresponding to no spike in this time interval and 1 in a time bin in which a spike has occurred. An example is illustrated in Fig. D.3. With small time bins we can assume a binary string with many more 0s than 1s, indeed many more than illustrated in the figure. We will also assume that the events in the bins are independent. Dependence would lower the entropy as mentioned before, and we are interested here in estimating the maximaum information carried in a spike train.

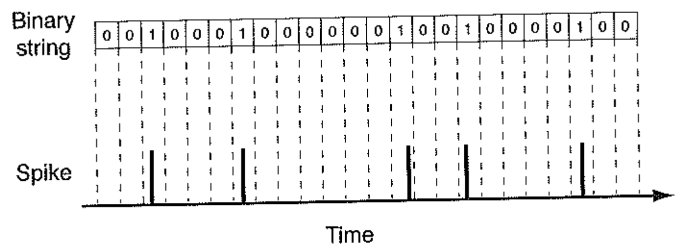

Fig. D.3 Representation of a spike train as a binary string with time resolution Δt.

The firing rate r of the spike train fixes the number of 1s and 0s in the spike train of length T. We can then calculate the number of possible spike patterns with the fixed number of spikes in the sequence of length T. The logarithm to base 2 of this number is the entropy of the spike train. This can thus be calculated to be

$$S = -\frac{T}{\Delta t \ln 2} \left[r\Delta t \ln (r\Delta t) + (1 - r\Delta t) \ln (1 - r\Delta t) \right]. \tag{D.12}$$

For time bins much smaller than the inverse firing rate, that is, $\Delta t \ll 1/r$, this is approximately equal to

$$S \approx Tr \log_2 \left(\frac{\mathrm{e}}{r\Delta t} \right). \tag{D.13}$$

The term $N = Tr$ is the number of spikes in the spike train, so that the average entropy per spike is given by

$$\frac{S}{N} \approx \log_2 \left(\frac{\mathrm{e}}{r\Delta t} \right). \tag{D.14}$$

This is the maximum entropy per spike of a spike train for a given timing precision Δt when we can assign to each individual spike pattern a unique message or sensory event. It is maximum in the sense that we considered only the noiseless case, and noise would diminish the information gain of each message.

The maximum entropy per spike is plotted for three different firing rates in Fig. D.4A. What is remarkable is that the entropy per spike is for the parameters shown always larger than one bit. The reason for this is that the interspike

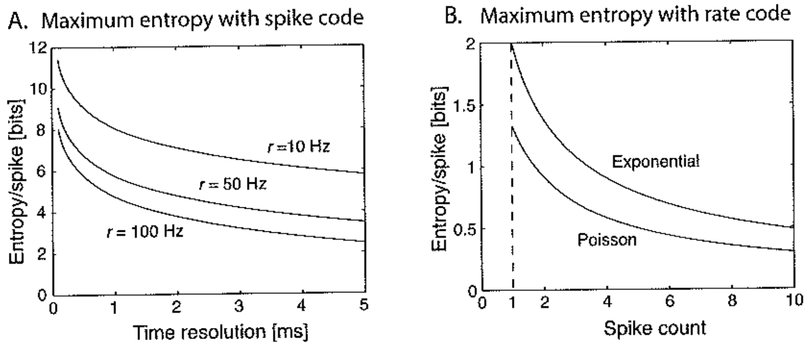

Fig. D.4 (A) Maximum entropy per spike for spike trains with different firing rates as a function of the size of the time bins, which sets the resolution of the firing times. (B) Maximum entropy per spike for spike trains from which the information is transmitted with a rate code. The two different curves correspond to two different spike statistics of the spike train, a Poisson and an exponential probability of spike counts. Spike trains with exponential spike distributions can convey the maximum information with a rate code for fixed variance.

intervals are long so that the occurrence of a spike becomes highly relevant. Also, we have normalized the entropy to the number of spikes, which can be a bit misleading; the absence of a spike can also bear information, and the measure of the entropy takes this into account. We have only normalized the spike train entropy to the total number of spikes to make it easier to compare the entropies of different spike trains and coding schemes.

Figure D.4A illustrates that the maximal entropy decreases with increasing precision interval Δt (decreasing time precision). This is because the number of patterns that can be represented by the spike train decreases with fewer time intervals in a spike train of length T. The entropy per spike also decreases with increasing firing rate because the individual spike is then not so surprising and bears less information. We should also keep in mind that the transmission of the information contained in a spike train is lowered by noise in the transmission process and by spike correlations. However, the maximum entropy as calculated above gives us the scale to which we can compare measurements of information transmission in the brain.

D.2.3 Entropy of a rate code

We can compare the entropy of a spike train above, in which we considered all possible spike patterns (which is therefore a time code), with other coding schemes that do not employ all possible spike patterns, such as a rate code. In the rate code only the number of spikes n in a certain interval T is relevant. The entropy of observing N spikes in a time interval T can be calculated by applying the definition of the entropy,

$$\begin{aligned} S(N;T) &= -\sum_n P(n)\log_2 P(n) \\ &= -\frac{1}{\ln 2}\sum_n P(n)\ln P(n) \end{aligned} \tag{D.15}$$

where $P(n)$ is the probability of observing n spikes in the time interval T. For example, let us calculate the entropy for a *Poisson spike train* of length T. The probability of observing n spikes, when we expect $N = rT$ spikes, is given by the Poisson distribution,

$$P(n) = \frac{N^n \mathrm{e}^{-N}}{n!}. \tag{D.16}$$

Inserting this probability into the log part of formula D.15 we get

$$S(N;T) = -\frac{1}{\ln 2}\sum_n P(n)(n \ln N - N - \ln(n!)). \tag{D.17}$$

We can approximate the last term with Stirling's formula

$$\ln(n!) \approx (n + \frac{1}{2})\ln n - n + \frac{1}{2}\ln(2\pi). \tag{D.18}$$

At this point we should remember the definition of the expectation value $E(x)$ of a quantity x,

$$E(x) = -\sum_x P(x)x, \tag{D.19}$$

and the expectation value of n is just $E(n) = N$. The entropy of a Poisson spike train with rate code is therefore

$$S(N;T) \approx \frac{1}{2}\log_2 N - \frac{1}{2}\log_2 2\pi. \tag{D.20}$$

We divide this value again by the number of spikes so that we get values for the entropy per spike. This is plotted in Fig. D.4B. The entropy of the rate code is, of course, much smaller than the entropy of a spike train in which we allowed the precise spike timing to be relevant. The length of time over which we have to integrate to achieve a particular spike count depends on the firing rate. To get a spike count of 2 we have to integrate on average for 40 ms if a neuron fires with mean firing rate of 50 Hz.

We calculated the entropy of a spike train with rate coding for the example of a Poisson spike train. Spike trains with other probability distributions of spike counts can have different entropies within the rate code. We can ask what would be the distribution of the spike count with fixed mean and variance that would result in the maximum entropy. The answer is that the entropy under these conditions is maximized by an exponential distribution. An exponentially distributed spike train

$$P(n) \propto \mathrm{e}^{-\lambda n} \tag{D.21}$$

with average firing rate $r = 1/[T(\exp(\lambda) - 1]$ has an entropy of

$$S(N;T) = \log_2(1 + N) + N\log_2(1 + \frac{1}{N}). \tag{D.22}$$

The entropy per spike for this distribution is also shown in Fig. D.4B. We can see that the entropy of such spike trains is moderately larger than the entropy for the Poisson spike train and reaches a value of precisely 2 bits/spike in the limit where a single spike carries all the information. The average information in a rate code can therefore be reasonably high so that the information that flows through the brain even with this code can be enormous.

D.3 Mutual information and channel capacity

So far we have only considered noise-free signal transmission and receivers that invert precisely the encoding of the transmitter ($g = f^{-1}$). This corresponds to perfect transmission and unambiguous decoding. However, in all practical applications of information transmission, including information transmission in neural systems, we must take noise into account. We have then to distinguish the received signal y from the sending signal x. The question we are most interested in is what a received signal y can tell us about an event x. Indeed, we will later attempt to reconstruct an event, such as a sensory input to the neural system, from the firing pattern of neurons. For now we want to ask, what the average information gain is from what a message y tells us about the set of events that could happen. We therefore have to consider conditional probabilities such as $P(x|y)$ (or corresponding density functions). The average information gain with such conditional probabilities over both sets is

$$\int_y P(y)S(X|y)\mathrm{d}y = \int_X \int_Y p(y)p(x|y) \log_2 \frac{p(x|y)}{p(x)} \mathrm{d}x\mathrm{d}y. \tag{D.23}$$

We can express the conditional probabilities with joint distributions,

$$P(x|y) = \frac{P(x,y)}{P(y)}, \tag{D.24}$$

so that we can also write eqn D.23 as

$$I^{\text{mutual}} = \int_X \int_Y p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)} \mathrm{d}x\mathrm{d}y. \tag{D.25}$$

This quantity is called the *cross-entropy* or *mutual information*. It is symmetrical in the arguments (which motivated the term mutual) and describes the average amount of information that can be gained by receiving a message y when a signal x is sent (or vice versa). We can also rewrite the mutual information as difference between the sum of the individual entropies $S(X)$ and $S(Y)$ and the entropy of the joint distributions $S(X,Y)$,

$$I^{\text{mutual}}(X,Y) = S(X) + S(Y) - S(X,Y). \tag{D.26}$$

The joint entropy in the case of independent events in X and Y is $S(X,Y) = S(X) + S(Y)$. The mutual information in this case is equal to zero,

$$I^{\text{mutual}}(\text{all } y \in Y \text{ independent of all } x \in X) = 0, \tag{D.27}$$

which reflects the fact that received messages are independent of sent messages. However, for information transmission we want messages such that the received message and the sent message are dependent. Mutual information not equal to zero then reflects some correlation of the events in X and Y. The mutual information in a communication channel describes the average amount of information we can expect to flow between input and output events.

An amount of information that can be transmitted by a change in the signal is proportional to the logarithm of the number of states the signal can have.

For example, in Fig. D.2A we have considered only two allowed positions of the thumb and could therefore send one bit of information in each time interval in which the thumb can change. We can increase the number of states the thumb can take as illustrated in Fig. D.5. There we allowed for three positions $(0°, 45°, 90°)$ and we could therefore transmit $I = \log_2 1/3 \approx 1.6$ bit of information at every time step with equally likely messages. We could increase the number of thumb states further until it will be difficult for the observer to distinguish the differences. There is therefore a limiting factor given by the resolution of our visual system and the noise introduced by the motor system of the sender. It is intuitively clear that the amount of information that can be transmitted over a noisy communication channel is limited by the ratio of the signal to noise or, to be more precise, by the ratios of the variance of these quantities, as the information is conveyed by changes in signals (see eqn D.10).

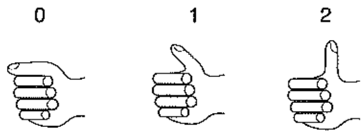

Fig. D.5 Example of hand signals using the thumb in three different states to transmit messages.

We can calculate the mutual information of a channel with noise for particular noise models and probability distributions for the input signal and the noise. It is common to consider additive Gaussian noise as well as Gaussian-distributed input signals. It turns out that the amount of information transmitted in such a Gaussian communication channel is the best we can achieve. The amount of information that can be transmitted through a Gaussian channel is called the *channel capacity*. The information we can transmit through a noisy channel is therefore always less than the channel capacity, which is given by

$$I \le \frac{1}{2} \log_2 \left(1 + \frac{\langle x^2 \rangle}{\langle \eta_{\text{eff}}^2 \rangle} \right), \tag{D.28}$$

where $\langle x^2 \rangle$ is the variance of the input signal and $\langle \eta_{\text{eff}}^2 \rangle$ is the effective variance of the channel noise, considering the noise as transmitted itself as signal through the channel. The ratio $\langle x^2 \rangle / \langle \eta_{\text{eff}}^2 \rangle$ is called the *signal to noise ratio* (SNR). It is interesting to note that the information transmission with given noise in the transmission channel can be increased by increasing the variability of the input signals. It seems that the high variability of spike trains, as discussed in Chapter 3, is therefore particularly suited for transmission in noisy neural systems. There are indeed several possible mechanisms that can increase the variability of neuronal response, some of which are discussed in Chapter 8.

D.4 Information and sparseness in the inferior-temporal cortex

D.4.1 Population information

To estimate the information content in a set of neurons, one has to record from several cells in an area simultaneously if the responses of the population neurons are not independent of each other. If, on the other hand, we assume that each neuron responds with a certain independent probability distribution to each stimulus, then we can use neurons that have been recorded independently. *Edmund Rolls* and colleagues have used such independent recordings from neurons in the inferior-temporal (IT) cortex to estimate the information in a population of neurons. The authors used neurons that responded to a set of 20 faces and used a statistical estimate for the decoding by fitting a Gaussian to the fluctuations of the neuronal response of each neuron. The results of the mutual information between the 20 face stimuli and the response of C recorded neurons is shown in Fig. D.6A with star symbols. The information quickly rises with the number of neurons taken into account in the analysis until the information gain levels off.

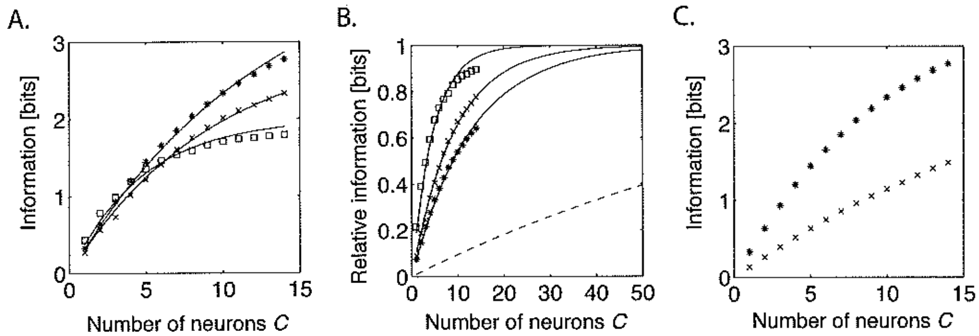

Fig. D.6 (A) Estimate of mutual information between face stimuli and firing rate responses of C cells in the inferior-temporal cortex. The set of stimuli consisted of 20 faces (stars), 8 faces (crosses), and 4 faces (squares). (B) The information in the population of cells relative to the number of stimuli in the stimulus set. The solid lines are fits of the cell data to the ceiling model (eqn D.29). The dashed line illustrates the values of the ceiling model for a stimulus set of 10,000 items and $y = 0.01$. (C) Estimate of mutual information with 20 faces when the neuronal response is derived from the spike count in 500 ms (stars) and 50 ms (crosses). [Data from Rolls, Treves, and Tovee, *Experimental Brain Research* 114: 149–62 (1997).]

The information for a stimulus set of 20 faces can be compared with the responses of the neurons to a reduced set of faces. The information, when analysed with a set of 8 faces (crosses) and a set of 4 faces (squares), is also shown in Fig. D.6A. We see that the levelling off of the curves depends on the size of the stimulus set, and the linear regime of the scaling extends further with increasing size of the stimulus set. The deviation from the linear envelope for larger numbers of neurons used for decoding can be expected because the entropy of a set of N equally likely objects is given by $S_{\max} = \log_2 N$. The entropy for a fixed size of the stimulus set cannot therefore continue to grow. The deviation of the linear scaling is therefore a *ceiling effect*. The data points are indeed well fitted with a simple empirical model describing the ceiling effect by

$$I(C) = S_{\max}(1 - (1 - y)^C), \tag{D.29}$$

where the parameter y is the fraction of information carried by each neuron. The fits reveal a decreasing fraction y of information carried by each neuron with increasing size of the stimulus set. This can be expected as each new stimulus that has to be represented by a neuron diminishes the ability to discriminate among of the other stimuli represented by the neuron.

On the basis of the simple ceiling effect model we can also plot the data of Fig. D.6A relative to the maximum entropy of the stimulus set. This is shown in Fig. D.6B. From this it seems that only a small number of neurons are necessary to accurately represent a stimulus set. This conclusion is, however,

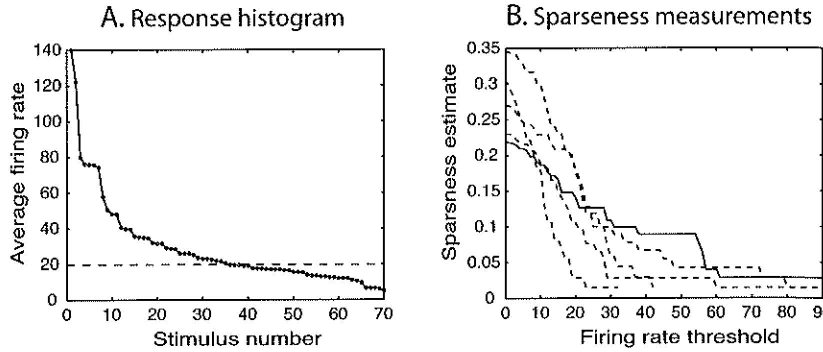

Fig. D.7 (A) Average firing rates of a neuron in the IT cortex in response to each individual stimulus in a set of 70 stimuli (faces and objects). The responses are sorted in descending order. The horizontal dashed line indicates the spontaneous firing rate of this neuron. (B) The sparseness derived from five IT neurons calculated from eqn 7.27 in which responses lower than the firing rate threshold above the spontaneous firing rate are ignored. The solid line corresponds to the data shown in (A). [Data from Rolls and Tovee, *Journal of Physiology* 73: 713–26 (1995).]

obscured by the use of small stimuli sets in the experiments. If we extrapolate the data using the ceiling model we find that much larger numbers of neurons are necessary to gain sufficient information about a presented stimulus. An example of a set size of 10,000 items with $y = 0.01$ is shown as a dashed line in Fig. D.6B. Larger set sizes with smaller y value would result in an even smaller increase of the information gain with the number of cells in the population. Also, the information gains in Fig. D.6A and B are overestimates if we think that the amount of information in shorter time intervals is relevant. In the experiments a 500 ms time interval following the stimulus was used to estimate the information gain. The estimates for a smaller time interval (50 ms) are shown in Fig. D.6C as crosses and compared to the information gain estimated with a 500 ms time interval (stars) from the previous analysis. The information gain therefore increases only slowly with the number of neurons in the population, which suggests a distributed representation of information.

D.4.2 Sparseness of object representations

We still have not answered the question of how sparse the representation in the IT cortex is. To do this we use the measured responses of individual neurons directly as input to eqn 7.27. The average firing rate of one particular IT neuron, averaged over many trials, in response to 70 individual stimuli is shown in Fig. D.7A. The responses are thereby ordered so that the stimuli with the largest responses are given a small stimulus number. We indicate the spontaneous firing rate of this neuron, against which we have to judge the response of the neuron, as a horizontal dashed line. The individual average firing rate response of a neuron to some stimuli are markedly different from the spontaneous firing of the neuron, whereas most such differences are only minor. If all such differences are relevant, then we see that the neuron responded to practically all stimuli so that we have to conclude that a very distributed code is used. However, if only large responses are relevant, then we would conclude a more sparse representation because only a small number of stimuli drive the cell with large responses.

To take the relevance in the firing rate into account we include in the following a firing rate threshold and count only cell responses above this threshold as relevant. The relevant firing rate of the neuron for a given stimulus is therefore the mean firing rate for responses above the threshold minus the spontaneous firing rate, and is zero for firing rates below the threshold. These relevant firing rates are used in eqn 7.27 to estimate the sparseness. The sparseness estimate from data from five different neurons as a function of the firing rate threshold are shown in Fig. D.7B. The solid line corresponds to the cell shown in Fig. D.7A. All cells indicate fairly small values of sparseness. A value of around 0.3 is only reached if we also take small deviations of the spontaneous firing rate into account as the relevant signal. The estimated sparseness, however, becomes much smaller if we take only high firing rates into account. It is also likely that this estimation of the sparseness in IT responses is an overestimation of the sparseness that we would get for much larger stimuli sets. The data set used in the experiments had several images that often drive responses in IT cells. The reported results indicate that sparsely distributed representations, roughly on the order of $a = 0.1$, may be employed in the brain.

D.5 Surprise

The above section reviewed classical concepts of information theory, in particular the average information (entropy) of different codes. When thinking about brain processing in such terms one must recognize that we are talking about a large number of possible states of the environment and the brain itself. Many of the concepts from classical information theory were defined for finite and even small communication codes. Also, it is useful to define information-theoretic measures about the average information gain of specific signals.

To define such a measure, we consider a system with a set of responses, r. Our knowledge about the system can be characterized by the *prior*, $P(r)$, for example by building a model that produces the characteristic distribution of responses. We then probe the system with a specific signal, s, and measure the responses to this specific signal. We can then update our knowledge of the system, which is the *posterior*, $P(r|s)$. If the response of the system is as expected, than we are not *surprised* about the response to this input and there is no need to update our model of the system. The posterior is then equal to the prior. However, the difference of the posterior is large for responses that surprise us. We can turn this observation around and define the *surprise* as the difference between the posterior and prior distribution. A common measure to quantify this difference is the Kulbach–Leibler divergence (see Appendix C.6), and the surprise is hence defined as

$$I^S(s) = \int_R P(r|s) \log_2 \frac{P(r|s)}{P(r)} \mathrm{d}r. \tag{D.30}$$

Another way of thinking about this measure is to write it as a difference

$$I^S(s) = -\int_R P(r|s)(\log_2 P(r) - \log_2 P(r|s) \mathrm{d}r. \tag{D.31}$$

The first term, $I(s) = \int_R P(r|s) \log_2 P(r) \mathrm{d}r$, is the total amount of information associated with each stimulus. However, not all the information is about the stimulus since some of the responses can also be expected generally from the system. We thus need to subtract the information not related to the stimulus, which can be measured from the variations when the stimulus is held fixed, $I(R|s) = \int_R P(r|s)\log_2 P(r|s)\mathrm{d}r$. This quantity defined as surprise looks like an information measure. However, this measure is based on conditional probabilities so that this quantity does not necessarily have the additivity feature that we demand from information measures.

Examples for the stimulus-dependent surprise per spike are shown in Fig. D.8 for two visual neurons in the inferior-temporal cortex that respond to faces. The average firing rate of the responding neurons shown in Fig. D.8A was 54 Hz. Responses with rates around this firing rate bear little surprise value. Some responses with high firing rates for this neuron carried high surprise values of around 2 bits per spike. However, the average surprise value in the responses of this neuron to all 20 faces in the stimulus set was only 0.55 bits. This value is not surprisingly high and does not suggest the need for a code within the firing pattern of these neurons. Another example is shown in Fig. D.8B. The surprise values are again lowest around the mean firing rates of the neuron, which is higher than that of the neuron shown on the left. This neuron carried the largest surprise values for small firing rates and carried an average surprise value similar to that in the previous example.

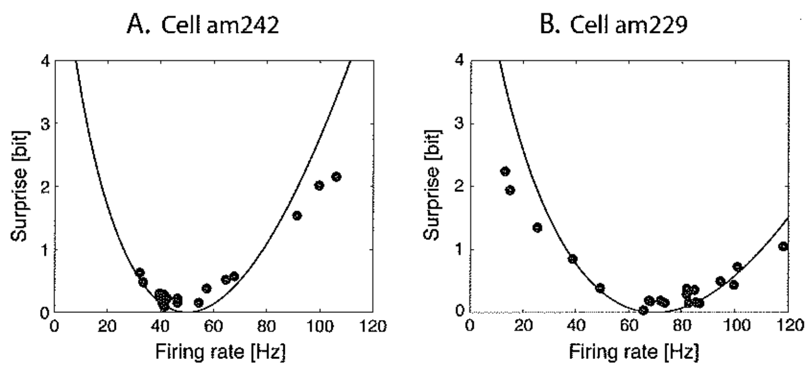

Fig. D.8 Stimulus-dependent surprise for two cells in the inferior-temporal cortex that respond to faces. The curve is 5 times the universal expectation for small time windows (eqn D.32). [Data from Rolls, Treves, Tovee, and Panzeri, *Journal of Computational Neuroscience* 4: 309–33 (1997).]

The values for the surprise rate were calculated for 500 ms spike trains during the presentation of the faces. Several studies have also shown that much of the information about a stimulus is transmitted in much smaller time intervals. For very small time intervals, so that at most one spike can occur during this time, there is not much freedom for a coding scheme, so that we can expect a universal curve, which is given by

$$I^s(\Delta t \ll 1) = r^s \log_2(\frac{r^s}{r}) + \frac{r - r^s}{\ln 2}, \tag{D.32}$$

where r^s is the specific response rate for stimulus s, and r is the average firing rate over all stimuli. This universal surprise curve, divided by the average firing rate r and multiplied by a factor of 5, is plotted in Fig. D.8 as a solid line. We scaled the curve up because the experimental data were obtained in large time windows for which the values can be higher. What is striking, however, is that the experimental data follow approximately the universal curve for small time windows. This is another hint that elaborate codes of spiking pattern might not be employed by neurons in the central nervous system.

A related formulation of surprise in a Bayesian view was proposed by *Pierre Baldi*, who also applied it to neuroscientific research with his colleague *Laurrent Itti*. In a Bayesian view, we consider a distribution of possible models that can explain the data. A new example, $\mathbf{x}$, can be used to update our world view by refining the distribution of models that can explain the data. Thus, we consider distributions of possible models, $P(\theta)$, where different models are specified with different parameters θ. The Bayesian surprise is defined as the change between the prior and posterior distribution of models,

$$S(\mathbf{x}) = \int_\theta P(\theta|\mathbf{x}) \log_2 \frac{P(\theta|\mathbf{x})}{P(\theta)} \mathrm{d}\theta. \tag{D.33}$$

New evidence is surprising when not anticipated by the system, and surprising data should thus trigger learning as discussed further in Chapter 10.

Further reading

The (two) classical articles by Claude Shannon (1948) are worth reading. A well known and wonderfully written book by Rieke *et al.* (1996) explores neural codes in nervous systems in much more detail and explains information theory on the way. The article by Allesandro Treves (2001) gives a good overview of research about the 'neural code' and is even recommended as an addition to the above book. The book chapter by Pierre Balidi also introduces surprise and is worth exploring, and his web page contains further papers where his concept of surprise is studied in neuroscience.

Claude Shannon (1948), *A mathematical theory of communication* in *Bell System Technical Journal*, 27.

Fred Rieke, David Warland, Rob de Ruyter van Steveninck, and William Bialek (1996), *Spikes: Exploring the neural code*, MIT Press.

Allesandro Treves (2001), *Information coding in higher sensory and memory areas*, in *Handbook of biological physics IV*, Stan Gielen (ed.), Elsevier.

Pierre Baldi (2002), *A computational theory of surprise*, in *Information, coding, and mathematics*, M. Blaum (ed.), 1–25, Kluwer.

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